



Azurite

△ **Sphere** of radiating azurite crystals

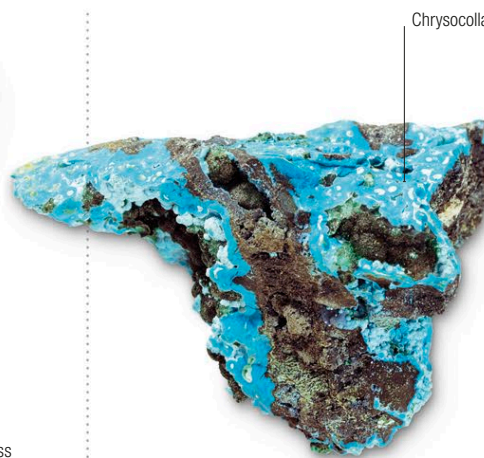
Azurite is thought to have been used in blue glaze in ancient Egypt, and it was used as a blue pigment in Renaissance European art. It takes its name from the Persian *lazhuward*, meaning “blue”. Azurite is cut *en cabochon* and, in rare cases, is faceted for collectors. Spheres of radiating azurite crystals more than 2.5cm (1in) in diameter are sometimes worn as jewellery, and slices of these are mounted in silver frames as pendants. Banded azurite and malachite used for ornamental purposes is called chessylite, after Chessy, France, where it was found.

Specification

Chemical name Copper carbonate | **Formula** $\text{Cu}_3(\text{CO}_3)_2(\text{OH})_2$
Colours Azure to dark blue | **Structure** Monoclinic | **Hardness** 3.5–4 | **SG** 3.7–3.9 | **RI** 1.72–1.85 | **Lustre** Vitreous to dull to earthy | **Streak** Blue | **Locations** France, Mexico, Australia, Chile, Russia, Morocco, Namibia, China



Large crystals | Rough | This stunning mixed specimen features a cluster of unusually large and finely crystallized azurites that have developed on a groundmass of goethite, a form of iron hydroxide.



Mixed minerals | Rough | Azurite often occurs with other copper minerals, as in this specimen of azurite and chrysocolla in a rock groundmass.



Australian azurite | Rough | Australia is a mineral-rich country, with various locations yielding azurite sources as well as extensive copper deposits.

Blue pigment

An alternative source of colour

Renaissance painters conventionally used lapis lazuli as a blue pigment. However, it was expensive, and azurite made a cheap and plentiful alternative. Unfortunately, powdered azurite is unstable in open air: moisture causes a replacement of some of the carbon dioxide in azurite with water, converting it to green malachite. This is why some of the blue in old paintings has turned green over time, including Giotto's fresco cycle in Padua, Italy, dating back to the early 14th century.



Giotto's Lamentation of Christ
Patches of azurite have deteriorated from their original blue colour.



Azurite heart | Cut | Azurite and malachite are commonly intermixed, and can make spectacular cabochons, as in this carved, heart-shaped stone.



Mixed cabochon | Cut | When cut in the proper direction, the patterns produced by mixtures of malachite and azurite can be spectacular, as demonstrated here.